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ACCELERATE YOUR GROWTH

Dev Ops with AI

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Accelerate Your Career Growth

www.thecloudtrain.com

CLOUD TRAIN
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Curriculum

What You'll Learn

1. INFRASTRUCTURE OVERVIEW
2. INTRODUCTION TO DEVOPS
3. VERSION CONTROL | GITHUB
4. CONTAINERIZATION | DOCKER – PART 1
5. CONTAINERIZATION | DOCKER – PART 2
6. CONFIGURATION MANAGEMENT | PUPPET
7. CONFIGURATION MANAGEMENT | ANSIBLE
8. CONTINUOUS TESTING | SELENIUM
9. CONTINUOUS INTEGRATION | JENKINS
10. CONTAINER ORCHESTRATION | KUBERNETES CONTINUOUS MONITORING | PROMETHEUS + GRAFANA
11. INFRASTRUCTURE AS A CODE | TERRAFORM

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Let's Begin

Infrastructure Overview

- AWS/GCP Clouds Walkthrough
- Linux Overview
- Installation of DevOps Tools in the Cloud
- Git + GitHub
- Docker
- Puppet
- Ansible
- Selenium
- Jenkins
- Kubernetes
- Prometheus + Grafana
- Terraform

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ANKIT KUMAR

★★★★★

It was an enriching workshop where I gained valuable insights into DevOps. The experience was truly invaluable, providing a comprehensive understanding of DevOps principles and practices

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PRATIKSHA

★★★★★

It would have been better for our learning if there was a group created where we could post our doubts so that the instructor as well as the other students could help clear our doubts.

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PRASHANT

★★★★★

Great explanations throughout the session! Please continue delivering the same high-quality insights in future trainings.

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SIVARAMAN

★★★★★

"The training and lab sessions exceeded my expectations! The hands-on experience was incredibly valuable, and the practical insights provided were spot on. It was an awesome and enriching experience 👍."

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VINAY

★★★★★

"It was incredibly informative and enlightening. I gained a wealth of knowledge and new insights."

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RAJASHEKAR

★★★★★

It's highly beneficial and practical, especially for beginners, providing a solid foundation in DevOps principles and techniques.

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ANIMESH

★★★★★

The workshop was excellent! I highly recommend starting a full-fledged training. I've decided to go for the lifetime membership.

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RAJESH

★★★★★

As a non-technical person, I couldn't take full advantage of the course which seems to be more suitable for technical resources.

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Module 1

Introduction to DevOps

- What is Software Development?
- Software Development Life Cycle
- Traditional Models for SDLC
- Agile Model
- Why DevOps?
- What is DevOps?
- DevOps Lifecycle
- DevOps Tools

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Module 2

Version Control | Git & GitHub

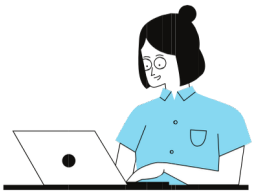
- What is Version Control?
- Types of Version Control Systems
- Introduction to SVN
- Introduction to Git
- Git Lifecycle
- How does Git work?
- Common Git Commands
- Working with Branches in Git
- Merging Branches
- Resolving Merge Conflicts
- Collaboration in GitHub
- GitHub Security
- Git Workflow
- GitHub Actions & Workflows

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Module 2

Version Control | Git/GitHub

Hands-on Lab



- Git life cycle commands
- Pushing code to GitHub
- Stashing code in Git Creating and deleting Git branches
- Reverting a push to GitHub
- Merging branches using git merge
- Merging branches using git rebase
- Resolving merge conflicts using the git merge tool

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Module 3

Containerization | Docker - Part 1

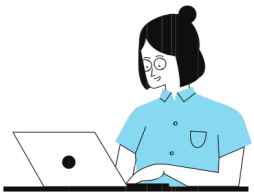
- Introduction to Virtualization
- Introduction to Containerization:
- Containerization Tools
- Introduction to Docker
- Understanding Docker Lifecycle
- Components of the Docker Ecosystem
- Installing Docker Common
- Docker Commands
- Creating a Docker Hub Account
- Committing Changes in a Pushing a Container
- Image to Docker Hub Dockerfile

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Module 3

Containerization | Docker - 1

Hands-on Lab



- Common Docker operations
- Creating a Docker Hub account Committing changes to a container
- Pushing the container to DockerHub
- Creating local image repository
- Building an image using a
- Dockerfile

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Module 4

Containerization | Docker - Part 2

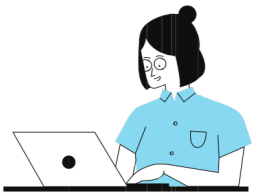
- What are Docker Volumes?
- Deploying a Multi-tier Application
- Microservices Using Docker Network
- YAML/JSON Using Docker Compose
- to Deploy Containers
- What is Container Orchestration?
- Container Orchestration Tools
- Introduction to Docker Swarm
- Deploying a 2-node Cluster Using Docker Swarm

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Module 4

Containerization | Docker - 2

Hands-on Lab



- Linking Containers
- Docker Volumes & Bind Mounts
- Creating Docker volumes Using Docker Compose to deploy multiple containers Deploying a
- multi-node cluster using Docker Swarm
- Deploying a multi-service app on Docker Swarm

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Module 5

Configuration Management | Puppet (Self Paced)

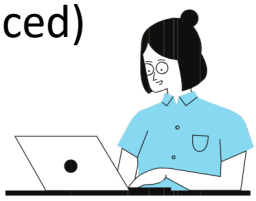
- Why Configuration Management?
- What is Configuration Management?
- Configuration Management Tools
- What is Puppet?
- Puppet Architecture
- Puppet Master–Slave Setup
- Code Basics for Puppet
- Applying Configuration Using Module

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Module 5

Configuration Management | Puppet

(Self Paced)



Hands-on Lab

- Puppet installation
- Implementing a Puppet manifest
- Implementing conditions in manifest
- Implementing loops in manifest
- Implementing variables in manifest
- Implementing a Puppet module

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KARISHMA

★★★★★

The trainer's quick and effective responses made for a seamless experience. Thank you, Cloud Train, for your excellent service!

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ABDURAOOF

★★★★★

The Training was great, and I learned valuable tools that will benefit my career. The Training video tutorials were especially helpful in making complex concepts easy to understand for future use.

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VIMAL KUMAR

★★★★★

The Training was fantastic – very detailed and hands-on, offering practical insights in real-world scenarios. The in-depth approach to each topic made it easier to grasp the concepts, and I feel much more confident in my skills now.

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SHREYA

★★★★★

This short and sweet Training is perfect for anyone new to DevOps and looking to gain a solid understanding of it easily. I really appreciated how CloudTrain consistently followed up with us throughout the Training and ensured all our doubts were cleared.

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SWAPNIL

★★★★★

Good. It would be beneficial to incorporate more practical-oriented guidance, especially in addressing various errors that may occur. This approach would enhance the learning experience and assist individuals in navigating challenges effectively.

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MANISH

★★★★★

I want to thank the organizers and instructors for delivering such an informative and well-structured Training. It was a valuable learning experience. The content was highly relevant and well-aligned with industry needs. The hands-on exercises on Jenkins, Docker, Ansible, and Kubernetes were particularly impactful, and the instructor's expertise and ability to simplify complex DevOps and cloud concepts were truly commendable. Thank you once again for organizing such an outstanding workshop.

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SHARIEF

★★★★★

I found the Cloud train to be incredibly insightful. The course materials were comprehensive, and the interactive sessions made learning enjoyable. Highly recommend it to anyone looking to improve their trading skills.

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NISAR

★★★★★

I recently attended the Cloud training DevOps class, which was a game-changer for me. The instructors were knowledgeable and the content was very well-structured. I feel much more confident in my skills now.

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Module 6

Configuration Management | Ansible

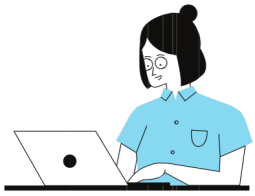
- What is Ansible?
- Ansible vs Puppet
- How Ansible Work?
- Ansible Architecture
- Setting up Master Slave Using Ansible
- Ansible Playbook
- Ansible Roles
- Applying Configuration Using Ansible
- Ansible Vault

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Module 6

Configuration Management | Ansible

Hands-on Lab



- Installing Ansible
- Creating a Playbook using YAML
- Variables/Conditions/Loops/
- Creating an Ansible roles Using
- roles in the playbook Ansible
- Vault

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Module 7

Continuous Testing | Selenium

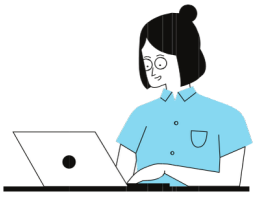
- Introduction to Software Testing
- Types of Software Testing
- Automated Testing Tools
- Introduction to Selenium
- Components of Selenium
- What is Maven?
- Creating Automated Tests
- Introduction to TestNG
- Introduction to Continuous Testing

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Module 7

Continuous Testing | Selenium

Hands-on Lab



- Setup java, eclipse and maven
- Creating selenium automated test in eclipse
- TestNG Annotations Web Scrapping

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Module 8

Continuous Integration | Jenkins

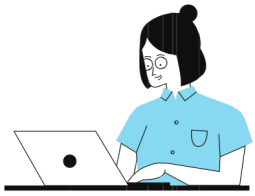
- Introduction to Continuous Integration
- What is Jenkins?
- Installing Jenkins
- Jenkins Master Slave Architecture
- Jenkins Integration with DevOps Tools
- Understanding CI/CD Pipelines
- Creating an End-to-end Automated CI/CD Pipeline

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Module 8

Continuous Integration | Jenkins

Hands-on Lab



- Creating a Jenkins Master Slave Setup Installing plug-ins in Jenkins
- Exploring Jenkins Configuration
- Creating Jenkins builds Creating
- scheduled builds Triggering jobs
- using Git web hooks Using the pipeline plugin in Jenkins

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Module 9

Container Orchestration | Kubernetes

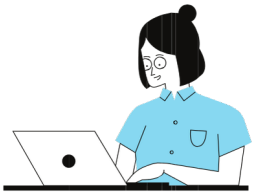
- Introduction to Kubernetes
- Docker Swarm vs Kubernetes
- Kubernetes Architecture
- Alternate Ways of Kubernetes Installation
- Deploying Kubernetes Using Kubeadm
- Deployment in Kubernetes
- Services in Kubernetes
- Ingress Controller in Kubernetes
- Ingress in Kubernetes
- Google Kubernetes Engine (GKE)

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Module 9

Container Orchestration | Kubernetes

Hands-on Lab



- Setting up Kubernetes using kubeadm Creating a deployment
- Creating services Installing ingress
- controller Creating an ingress
- Demonstrating the use of ingress,
- services and deployments together
- GKE Cluster Setup

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Module 10

Continuous Monitoring |

Prometheus + Grafana

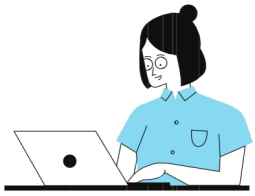
- What is Prometheus
- Prometheus architecture
- Prometheus installation & configuration
- Monitoring nodes with exporters
- Prometheus metrics querying
- Alerting with Prometheus
- Client libraries
- Grafana installation & configuration

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Module 10

Continuous Monitoring |

Prometheus + Grafana



Hands-on Lab

- Installing and Configuring Prometheus
- Querying Metrics
- Setting up Alerts
- Installing and Configuring Grafana

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Module 11

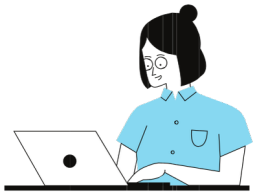
Infrastructure as a Code | Terraform

- What is Infrastructure-as-Code?
- IaC vs Configuration Management
- Introduction to Terraform
- Installing Terraform
- Basic Operations in Terraform – Init | Plan | Apply | Destroy
- Terraform Code Basics
- Deploying an End-to-end Architecture on GCP Using Terraform

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Module 11

Infrastructure as a Code | Terraform



Hands-on Lab

- Installing Terraform
- Initializing Terraform Provider Creating a Compute Instance using Terraform Updating changes to
- Compute Instance using Terraform
- Destroying Compute Instance using
- Terraform

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Module 12

DevOps with AI

- Overview of Generative Al
- Traditional VS Generative Al
- Some Use Cases
- Large Language Models
  - What is LargeLanguageModels (LLM)?
  - How to interactwithLIM?
- Al Landscape for Devops
  - AIChatBotsChatGPT,llama, Deepseek, Gemini
  - Al Assistants Github Co-pilot Workspace, Pieces for developers
  - Al Agents GitHub Co-pilot workspace, Bolt.new
  - Programming Pythons, Django, FastApi

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Module 12

DevOps with AI

- Difference between AI ChatBots, Assistants, & Agents
  - AITools forDevOps
  - GitHub Co-pilot
  - ChatGPT
  - Llama
  - Kubiya (Al for DevOps)
  - Snyk (Al for Security)
  - Otter (Al for Documentation)
  - Eraser (Al for Documentation)
- Overview of AIOps
  - WhatisAlops,and how does it work?
  - AI-powered incident detection and auto-remediation
  - Using Al for predictive maintenance and anomaly detection
  - Tools: Dynatrace, Moogsoft, IBM Watson AIOps

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Module 12

DevOps with AI



Hands-on Lab

- Demo on LLMS
- AI for CICD pipelines Generate a Github actions pipelines or Jenkins pipeline using AI.
- Al for Kubernetes Generate Kubernetes App deployment manifest including service, ingresses, etc
- Al for Documentation Small demo with Otter and Eraser
- Capstone Project Implement an Al-assisted End to End DevOps Automation

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VINAY

★★★★★

"It was incredibly informative and enlightening. I gained a wealth of knowledge and new insights."

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RAJASHEKAR

★★★★★

It's highly beneficial and practical, especially for beginners, providing a solid foundation in DevOps principles and techniques.

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MEGHANA

★★★★★

It was a great learning, easy to understand to concepts with good explanation in each session. I would recommend strongly.

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ABHIRAM

★★★★★

It's really good and affordable. Please keep up this work as it will be very useful to many

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CHETHAN

★★★★★

I am happy that I joined the course and understood where I stand which helped me to plan for the next year to coupe my skills. Please update the documentation and please mention or document on how to create Ubuntu machine at start to avoid confusion

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RAGHAV

★★★★★

I had a blast in this workshop! I gained valuable knowledge and grasped everything effortlessly. I've already referred my friends to join your upcoming sessions, praising your exceptional teaching skills. Thank you so much sir

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SAGNIK

★★★★★

Your content provides a concise yet highly valuable overview of the DevOps process. It's ideal for individuals seeking insights without delving into the nitty-gritty details necessary for hands-on work.

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MANIKANTESWARA

★★★★★

Your explanation on each topic is incredibly informative and precise, making it highly valuable. Your insights are truly worth the time and attention.

Accelerate your career growth

We can help you accelerate your career growth to move ahead faster and achieve your professional goals quicker.

Whether

- You're already having years of experience in the industry OR
- You're coming from a different industry & want to switch jobprofileOR
- You're just getting started. ***..Letushelpyou.***

Got Queries Or need more info?

Contact us

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For questions and more details, please visit https://www.thecloudtrain.com/, or

email @ join@thecloudtrain.com

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